1. Introduction to Finite Element Method

1.1 What is FEM?

- The Finite Element Method (FEM) is a numerical technique used to find approximate solutions of boundary value problems in engineering.
- The whole structure (continuum) is divided into a finite number of small pieces called finite elements, connected at points called nodes.
- Each element has a simple behaviour described by a displacement (shape) function; the behaviour of the whole body is obtained by assembling all elements.
- Result: a large set of simultaneous algebraic equations $[K]\{u\} = \{F\}$ which is solved by computer.

1.2 Brief history

- 1943 Courant introduced the idea using triangular elements for torsion problems.
- 1956 Turner, Clough, Martin and Topp used FEM for aircraft structural analysis (stiffness method).
- 1960 Clough coined the term "Finite Element Method".
- Later extended to heat transfer, fluid flow, electromagnetics, vibration, etc.

1.3 General steps in FEM

- Step 1: Discretise the structure (divide into elements and nodes).
- Step 2: Select a suitable displacement function (shape function).
- Step 3: Derive the strain-displacement and stress-strain relations.
- Step 4: Derive the element stiffness matrix and load vector.
- Step 5: Assemble element equations to get the global equation $[K]\{u\}=\{F\}$.
- Step 6: Apply boundary conditions (supports and loads).

- Step 7: Solve for the nodal displacements.
- Step 8: Calculate element strains and stresses (post-processing).

1.4 Applications of FEM

- Structural analysis: beams, trusses, frames, plates, shells, bridges, dams.
- Mechanical: stress concentration, machine parts, automobile and aircraft bodies.
- Thermal analysis: conduction and convection in solids.
- Fluid flow, seepage through soil, electromagnetic fields, vibration and buckling.

1.5 Advantages of FEM

- Can handle complex geometry, irregular boundaries and any shape of body.
- Can handle different material properties, anisotropic and non-homogeneous materials.
- Different types of boundary conditions and loads can be applied easily.
- The same general program can be used for many different problems.
- Element size can be varied: fine mesh where stress changes rapidly, coarse mesh elsewhere.
- Dynamic, non-linear and time dependent problems can also be solved.
- Saves cost of prototypes; the design can be modified quickly on the computer.

1.6 Disadvantages of FEM

- Requires a computer and good, costly software; large memory and computing time.
- The result is only approximate, not exact; accuracy depends on the mesh.
- Skill is needed to prepare the model; wrong inputs give wrong results (garbage in, garbage out).

- Large amount of input data and output data to be checked.
- Stress values at nodes may show errors (discontinuity of stress between elements).
- Not economical for very simple problems where a closed form solution is available.

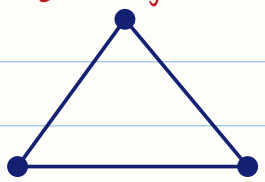
FEM = approximate method; Exact solution (theory of elasticity) is not possible for complex shapes.

1-D

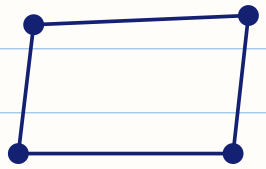


line / bar / beam

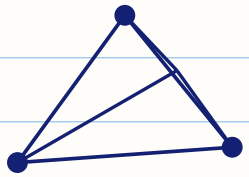
2-D triangle



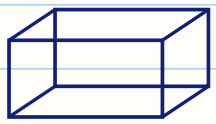
2-D quad



3-D tetrahedron



3-D hexahedron (brick)



2. Basic Equations in Elasticity

2.1 Stress at a point

- Stress = internal resisting force per unit area. Normal stress σ (perpendicular to the face) and shear stress τ (parallel to the face).
- In 3-D, six independent stress components act at a point:

$$\{\sigma\} = \{\sigma_x \; \sigma_y \; \sigma_z \; \tau_{xy} \; \tau_{yz} \; \tau_{zx}\}^T$$

- Moment equilibrium of an element gives $\tau_{xy} = \tau_{yx}$, $\tau_{yz} = \tau_{zy}$, $\tau_{zx} = \tau_{xz}$.

2.2 Strain at a point

- Strain = change in dimension per unit original dimension. Normal strain ε and shear strain γ.

$$\{\varepsilon\} = \{\varepsilon_x\ \varepsilon_y\ \varepsilon_z\ \gamma_{xy}\ \gamma_{yz}\ \gamma_{zx}\}^T$$

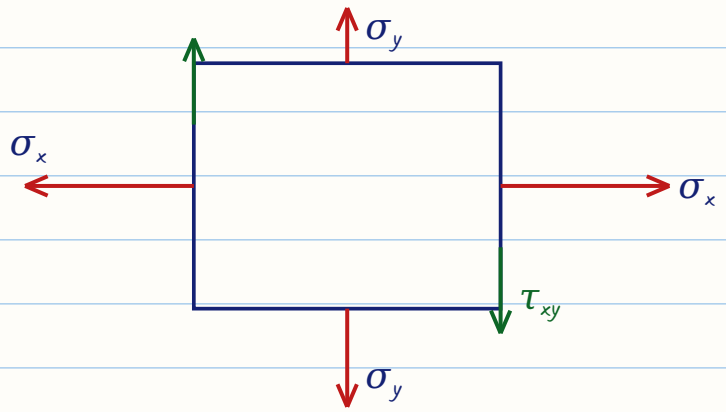
- If u, v, w are the displacements in the x, y, z directions, then the strain-displacement relations are:

$$\varepsilon_x = \partial u/\partial x,\ \varepsilon_y = \partial v/\partial y,\ \varepsilon_z = \partial w/\partial z$$

$$\gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

$$\gamma_{yz} = \partial v/\partial z + \partial w/\partial y$$

$$\gamma_{zx} = \partial w/\partial x + \partial u/\partial z$$



2-D stress element

$$\sigma = \{\sigma_x\ \sigma_y\ \tau_{xy}\}^T$$

$$\varepsilon = \{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T$$

$\tau_{xy} = \tau_{yx}$ (complementary shear)

2.3 Equilibrium equations (3-D)

- For an element in equilibrium (body forces X, Y, Z per unit volume):

$$\partial\sigma_x/\partial x + \partial\tau_{xy}/\partial y + \partial\tau_{xz}/\partial z + X = 0$$

$$\partial\tau_{xy}/\partial x + \partial\sigma_y/\partial y + \partial\tau_{yz}/\partial z + Y = 0$$

$$\partial\tau_{xz}/\partial x + \partial\tau_{yz}/\partial y + \partial\sigma_z/\partial z + Z = 0$$

2.4 Stress – strain relation (Hooke's law)

- For a linear elastic, isotropic material, with E = Young's modulus, ν = Poisson's ratio, G = shear modulus:

$$\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)] / E$$

$$\varepsilon_y = [\sigma_y - \nu(\sigma_z + \sigma_x)] / E$$

$$\varepsilon_z = [\sigma_z - \nu(\sigma_x + \sigma_y)] / E$$

$$\gamma_{xy} = \tau_{xy}/G,\ \gamma_{yz} = \tau_{yz}/G,\ \gamma_{zx} = \tau_{zx}/G,\ G = E / 2(1+\nu)$$

- In matrix form $\{\sigma\} = [D]\{\varepsilon\}$ where $[D]$ is the elasticity (material) matrix.

2.5 [D] matrix for 3-D (isotropic)

$$[D] = E(1-\nu) / [(1+\nu)(1-2\nu)] \times [\text{matrix}]$$

- The 6×6 matrix has: diagonal terms for normal stress = 1, coupling terms = $\nu/(1-\nu)$, shear diagonal terms = $(1-2\nu) / 2(1-\nu)$; all others are zero.

3-D problems need 6×6 [D]. Plane problems reduce the size to 3×3.

3. Concept of Plane Stress

3.1 Definition

- Plane stress occurs in a THIN plate loaded only in its own plane (in-plane loads), with thickness t very small compared to the other dimensions.
- Stresses in the z-direction (thickness) are zero:

$$\sigma_z = 0,\ \tau_{xz} = 0,\ \tau_{yz} = 0$$

- Non-zero stresses: σ_x, σ_y, τ_{xy}. But strain $\varepsilon_z \neq 0$ (the plate becomes thinner or thicker).
- Examples: thin plate with a hole, gear tooth (thin), deep beam, thin disc, plate under in-plane tension.

3.2 Stress-strain relation for plane stress

$$\varepsilon_x = (\sigma_x - \nu\,\sigma_y) / E$$

$$\varepsilon_y = (\sigma_y - \nu\,\sigma_x) / E$$

$$\gamma_{xy} = \tau_{xy} / G = 2(1+\nu)\,\tau_{xy} / E$$

- Solving for the stresses, $\{\sigma\} = [D]\{\varepsilon\}$:

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = E/(1-\nu^2)\,[\,1\ \nu\ 0\,;\,\nu\ 1\ 0\,;\,0\ 0\ (1-\nu)/2\,]\,\{\varepsilon_x\ \varepsilon_y\ \gamma_{xy}\}^T$$

Plane stress $[D] = E / (1-\nu^2) \times [\,1\ \nu\ 0 / \nu\ 1\ 0 / 0\ 0\ (1-\nu)/2\,]$

- Out-of-plane strain: $\varepsilon_z = -\nu\,(\sigma_x + \sigma_y) / E$

4. Concept of Plane Strain

4.1 Definition

- Plane strain occurs in a LONG body whose cross-section and loading do not change along the length (z-axis), and the ends are restrained.
- Strains in the z-direction are zero:

$$\varepsilon_z = 0,\ \gamma_{xz} = 0,\ \gamma_{yz} = 0$$

- Non-zero strains: ε_x, ε_y, γ_{xy}. But $\sigma_z \neq 0$ ($\sigma_z = \nu(\sigma_x + \sigma_y)$).
- Examples: retaining wall, long dam, tunnel, pipe under internal pressure, rolling cylinder, foundation under strip footing.

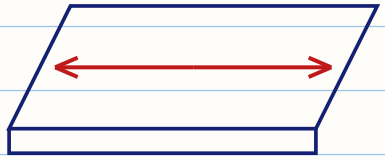
4.2 Stress-strain relation for plane strain

$$\varepsilon_x = (1+\nu)/E\,[\,(1-\nu)\sigma_x - \nu\,\sigma_y\,]$$

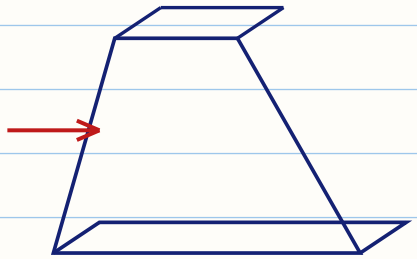
$$\varepsilon_y = (1+\nu)/E\,[\,(1-\nu)\sigma_y - \nu\,\sigma_x\,]$$

$$\{\sigma_x\ \sigma_y\ \tau_{xy}\}^T = E(1-\nu)\,/\,[(1+\nu)(1-2\nu)]\,[\,1\ \nu/(1-\nu)\ 0\,;\,\nu/(1-\nu)\ 1\ 0\,;\,0\ 0\ (1-2\nu)/2(1-\nu)\,]\,\{\varepsilon\}$$

Plane strain $[D] = E / [(1+\nu)(1-2\nu)] \times [\,1-\nu\ \nu\ 0 / \nu\ 1-\nu\ 0 / 0\ 0\ (1-2\nu)/2\,]$



Plane stress: thin plate
t small, $\sigma_z=\tau_{xz}=\tau_{yz}=0$



Plane strain: long body
$\varepsilon_z=\gamma_{xz}=\gamma_{yz}=0$

4.3 Plane stress vs plane strain

- Plane stress: $\sigma_z = 0, \varepsilon_z \neq 0$. Plane strain: $\varepsilon_z = 0, \sigma_z \neq 0$.
- Plane stress is for thin bodies; plane strain is for thick/long bodies.
- To change a plane stress [D] into plane strain [D], replace E by $E/(1-\nu^2)$ and ν by $\nu/(1-\nu)$.
- Both reduce a 3-D problem to a 2-D problem, with only three stress and three strain components.

5. Element Shapes, Nodes and Degrees of Freedom

5.1 Finite element

- A finite element is a small, simple sub-region of the structure. All elements together form the "mesh".
- Element shapes depend on the dimension of the problem:
- 1-D elements: straight or curved line (bar, truss, beam, spring) with 2 or 3 nodes.
- 2-D elements: triangle (3 or 6 nodes), quadrilateral (4, 8 or 9 nodes) – plane stress, plane strain, plates, shells.
- 3-D elements: tetrahedron (4 or 10 nodes), hexahedron/brick (8 or 20

nodes), wedge.

- Triangles fit curved and irregular boundaries well; quadrilaterals give better accuracy per element.

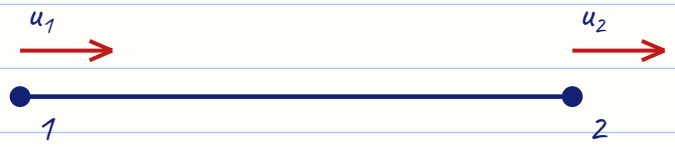
5.2 Nodes

- Nodes are the specific points of an element where the unknowns (displacements) are calculated. Elements are connected to each other only at nodes.
- Exterior (corner) nodes define the geometry. Interior / mid-side nodes are added for higher order elements.
- Nodes should be placed at points of load application, supports, change in thickness / material / geometry.

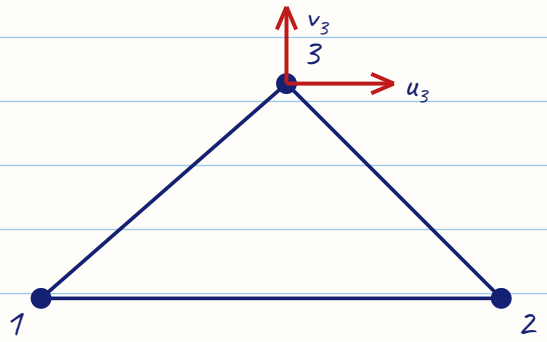
5.3 Nodal degrees of freedom (dof)

- Degrees of freedom = number of independent displacements (or rotations) of a node.
- Bar / truss (1-D): 1 dof per node (u). Plane truss: 2 dof per node (u, v).
- Beam element: 2 dof per node (deflection v and slope θ).
- 2-D plane stress / strain element: 2 dof per node (u, v).
- 3-D solid element: 3 dof per node (u, v, w). Frame / shell: up to 6 dof per node.

  Total dof of element = (number of nodes) × (dof per node)
- Example: CST has 3 nodes × 2 = 6 dof. 8-node brick has 8 × 3 = 24 dof.



1-D bar: 1 dof per node



2-D: 2 dof per node (u, v)

6. Displacement Function (Shape Function)

6.1 Concept

- The actual displacement inside an element is unknown, so we ASSUME a simple function of position, usually a polynomial. It is called the displacement function.
- The polynomial must give the displacement at every point inside the element in terms of the nodal displacements.
- Polynomial is preferred because it is easy to differentiate and integrate.

1-D bar: $u = a_1 + a_2x$ (2 constants for 2 nodes)

2-D triangle: $u = a_1 + a_2x + a_3y$, $v = a_4 + a_5x + a_6y$

2-D 4-node quad: $u = a_1 + a_2x + a_3y + a_4xy$

Number of terms in polynomial = number of dof of the element.

6.2 Writing in terms of shape functions

$$\{u\} = [N]\{q\}$$

- $\{u\}$ = displacement at any point, $[N]$ = matrix of shape functions, $\{q\}$ = nodal displacement vector.
- Properties of shape functions:
- $N_i = 1$ at node i and $N_i = 0$ at all other nodes.

- Sum of all shape functions = 1 at every point (rigid body motion is represented).
- Shape functions are continuous within the element.

6.3 Convergence requirements of displacement function

- Compatibility: displacement must be continuous across element boundaries (no gaps or overlaps).
- Completeness: the function must include rigid body motion and constant strain states.
- The polynomial should be complete up to its order, and geometrically isotropic (no preferred direction) – use Pascal's triangle.

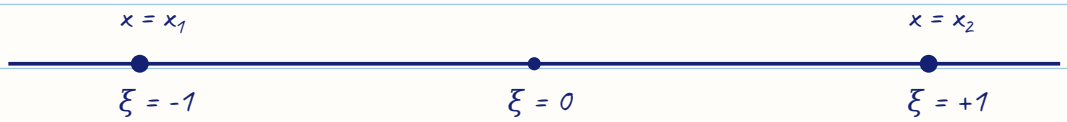
Pascal's triangle : 1 ; x y ; x^2 xy y^2 ; x^3 x^2y xy^2 y^3 ...

7. Natural Coordinates

7.1 Meaning

- A natural (local) coordinate system is attached to the element itself. Its value lies between -1 and +1 (or 0 and 1) and does not depend on the actual size or position of the element.
- Shape functions are written in natural coordinates, which makes the formulation general and suitable for numerical integration.
- Types: 1-D natural coordinate ξ, area (triangular) coordinates L_1, L_2, L_3, and volume coordinates for a tetrahedron.

7.2 1-D natural coordinate



$\xi = 2(x - x_1) / (x_2 - x_1) - 1$

$N_1 = (1-\xi)/2, \quad N_2 = (1+\xi)/2$

$x = N_1x_1 + N_2x_2,\ N_1 = (1-\xi)/2,\ N_2 = (1+\xi)/2$

- Jacobian $J = dx/d\xi = (x_2-x_1)/2 = l/2$, so $dx = (l/2)\, d\xi$.

7.3 Area coordinates for a triangle

- A point P inside a triangle 1-2-3 divides it into three small areas A_1, A_2, A_3.

  $L_1 = A_1/A,\ L_2 = A_2/A,\ L_3 = A_3/A$, A = total area

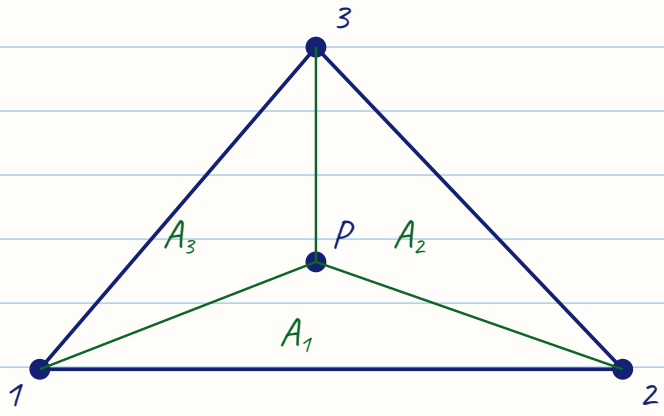
  $L_1 + L_2 + L_3 = 1$ (only two are independent)

- Linear shape functions of a triangle are equal to the area coordinates: $N_1 = L_1$, $N_2 = L_2$, $N_3 = L_3$.

  $x = L_1x_1 + L_2x_2 + L_3x_3,\ y = L_1y_1 + L_2y_2 + L_3y_3$

- Integration formula (very useful):

  $\int L_1^a L_2^b L_3^c\, dA = [a!\, b!\, c! / (a+b+c+2)!] \times 2A$



$L_i = A_i / A$

$L_1+L_2+L_3 = 1$

$L_i = 1$ at node i,

$L_i = 0$ on opposite side

8. Strain – Displacement Relations

8.1 For 1-D bar element

- With $u = N_1u_1 + N_2u_2$, the axial strain is:

  $\varepsilon = du/dx = (u_2 - u_1) / l$

  $\{\varepsilon\} = [B]\{q\}$, $[B] = [\,-1/l \;\; 1/l\,]$
- Stress $\sigma = E\,\varepsilon = E\,[B]\{q\}$.

8.2 For 2-D element

- Strains are found by differentiating the assumed displacements:

  $\{\varepsilon\} = \{\varepsilon_x \; \varepsilon_y \; \gamma_{xy}\}^T = \{\partial u/\partial x \; \partial v/\partial y \; \partial u/\partial y + \partial v/\partial x\}^T$
- Writing $u = [N]\{q\}$:

  $\{\varepsilon\} = [B]\{q\}$
- $[B]$ is the strain-displacement matrix; its terms are derivatives of the shape functions.
- Element stress:

  $\{\sigma\} = [D]\{\varepsilon\} = [D][B]\{q\}$
- Element stiffness matrix (thickness t, volume V):

  $[k] = \int [B]^T [D] [B] \, dV = t \int [B]^T [D][B] \, dA$

For a triangle with linear shape functions, $[B]$ is constant; therefore strain and stress are constant over the element (CST).

8.3 Important formulae to remember

- Strain energy $U = \tfrac{1}{2} \int \{\varepsilon\}^T \{\sigma\} \, dV = \tfrac{1}{2} \{q\}^T [k] \{q\}$
- Potential energy $\Pi = U - W$ (W = work done by external loads)
- Principle of minimum potential energy: $\partial\Pi/\partial q = 0$ gives $[k]\{q\} = \{F\}$.

Short questions: (1) define node, (2) state any four advantages of FEM, (3)

differentiate plane stress and plane strain, (4) write the [D] matrix for plane stress, (5) what is a shape function?